

APPENDIX D

Hook Events Reference

All 33 events with what fires them and what the matcher selects, then the exit-code and JSON output contract. Chapter 10 has worked examples.

Generated from the official docs snapshot of 2026-09-10 (Claude Code v2.1.267) by tools/gen_appendices.py. Do not edit by hand; rerun the script for the next edition.

EVENT	FIRES WHEN	MATCHER
SessionStart	session begins or resumes	how session started (startup, resume, clear, compact, fork)
Setup	--init-only, or --init/--maintenance in -p mode	CLI flag (init, maintenance)
UserPromptSubmit	prompt submitted, before Claude processes it	none
UserPromptExpansion	user-typed command expands into prompt	command name
PreToolUse	before tool call executes	tool name
PermissionRequest	tool call needs permission decision	tool name
PermissionDenied	auto mode denies tool call	tool name

EVENT	FIRES WHEN	MATCHER
PostToolUse	after tool call succeeds	tool name
PostToolUseFailure	after tool call fails	tool name
PostToolBatch	after batch of parallel tool calls resolves	none
Notification	Claude Code sends a notification	notification type
MessageDisplay	while assistant message text streams to screen	none
SubagentStart	subagent spawned	agent type
SubagentStop	subagent finishes	agent type
TaskCreated	task created via TaskCreate	none
TaskCompleted	task marked completed	none
Stop	Claude finishes responding	none
StopFailure	turn ends due to API error	error type (cloud_credential_error needs v2.1.267+)
TeammateIdle	agent team teammate about to go idle	none
InstructionsLoaded	CLAUDE.md or rules file loaded into context	load reason
ConfigChange	configuration file changes during session	configuration source
CwdChanged	working directory changes	none
DirectoryAdded	directory added via /add-dir or register_repo_root	how added (slash_command, register_repo_root)
FileChanged	watched file changes on disk	literal filenames to watch
WorktreeCreate	worktree being created; replaces git default	none

EVENT	FIRES WHEN	MATCHER
WorktreeRemove	worktree being removed	none
PreCompact	before context compaction	compaction trigger (manual, auto)
PostCompact	after compaction completes	compaction trigger (manual, auto)
PreModelSwitch	before requested model switch; can block	canonical target model name
PostModelSwitch	after session model changes	canonical target model name
Elicitation	MCP server requests user input	MCP server name
ElicitationResult	after user responds to MCP elicitation	MCP server name
SessionEnd	session terminates	exit reason

Handler types, exit codes, and output fields

- Hook handler types: `command`, `http`, `mcp_tool`, `prompt`, `agent` [agent experimental]
- Exit code 0: success; JSON on stdout parsed; plain stdout context on some events
- Exit code 2: blocking error; `stderr` is reason; JSON `allow` can't override
- Other exit codes: non-blocking unless valid JSON decides; `WorktreeCreate` fails on any nonzero
- Universal JSON fields: `continue`, `stopReason`, `suppressOutput`, `systemMessage`, `terminalSequence`
- Top-level decision fields: `decision` ("block"), `reason`
- `hookSpecificOutput` requires `hookEventName`
- `hookSpecificOutput.permissionDecision`: `allow`, `deny`, `ask`, `defer` (`PreToolUse`); `allow`, `deny`, `ask` (`PreModelSwitch`)
- `hookSpecificOutput.permissionDecisionReason`
- `hookSpecificOutput.updatedInput` (`PreToolUse`); `decision.updatedInput` (`PermissionRequest`)

- `hookSpecificOutput.additionalContext`
- `hookSpecificOutput.decision.behavior` (allow/deny), `decision.updatedPermissions`, `decision.message`, `decision.interrupt` (`PermissionRequest`)
- `hookSpecificOutput.retry` (`PermissionDenied`)
- `hookSpecificOutput.updatedToolOutput`, `updatedMCPToolOutput`, `classifierContext` (`PostToolUse`)
- `hookSpecificOutput.worktreePath` (`WorktreeCreate HTTP`)
- `hookSpecificOutput.action`, `content` (`Elicitation`, `ElicitationResult`)
- `hookSpecificOutput.displayContent` (`MessageDisplay`)
- `hookSpecificOutput.initialUserMessage`, `sessionTitle`, `watchPaths`, `reloadSkills` (`SessionStart`)
- `hookSpecificOutput.sessionTitle`, `suppressOriginalPrompt` (`UserPromptSubmit`)
- `watchPaths` (`CwdChanged`, `FileChanged`)
- **Prompt/agent hook response:** ok, reason, impossible (prompt only)
- **Permission update entry types:** `addRules`, `replaceRules`, `removeRules`, `setMode`, `addDirectories`, `removeDirectories`
- **Permission update destination:** `session`, `localSettings`, `projectSettings`, `userSettings`